



System Version Description
for the
The Balance Project
DCN: SVD20231222-P1:128
Revision Date: 05 Dec, 2025

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Controlled by: AGENCY
Controlled by: OFFICE
CUI Category: WILL-BE
Limited Dissemination Control: TBD
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Distribution Statement D: Vinay Agarwal



DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Project artifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1



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CHAPTER 1

Scope

This document provides the System Version Description (**SVD**) for the Balance System. The system will be referred to as the Balance System.

This document is generally cited as [ref`KNEAD`SVD`ExProj].

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

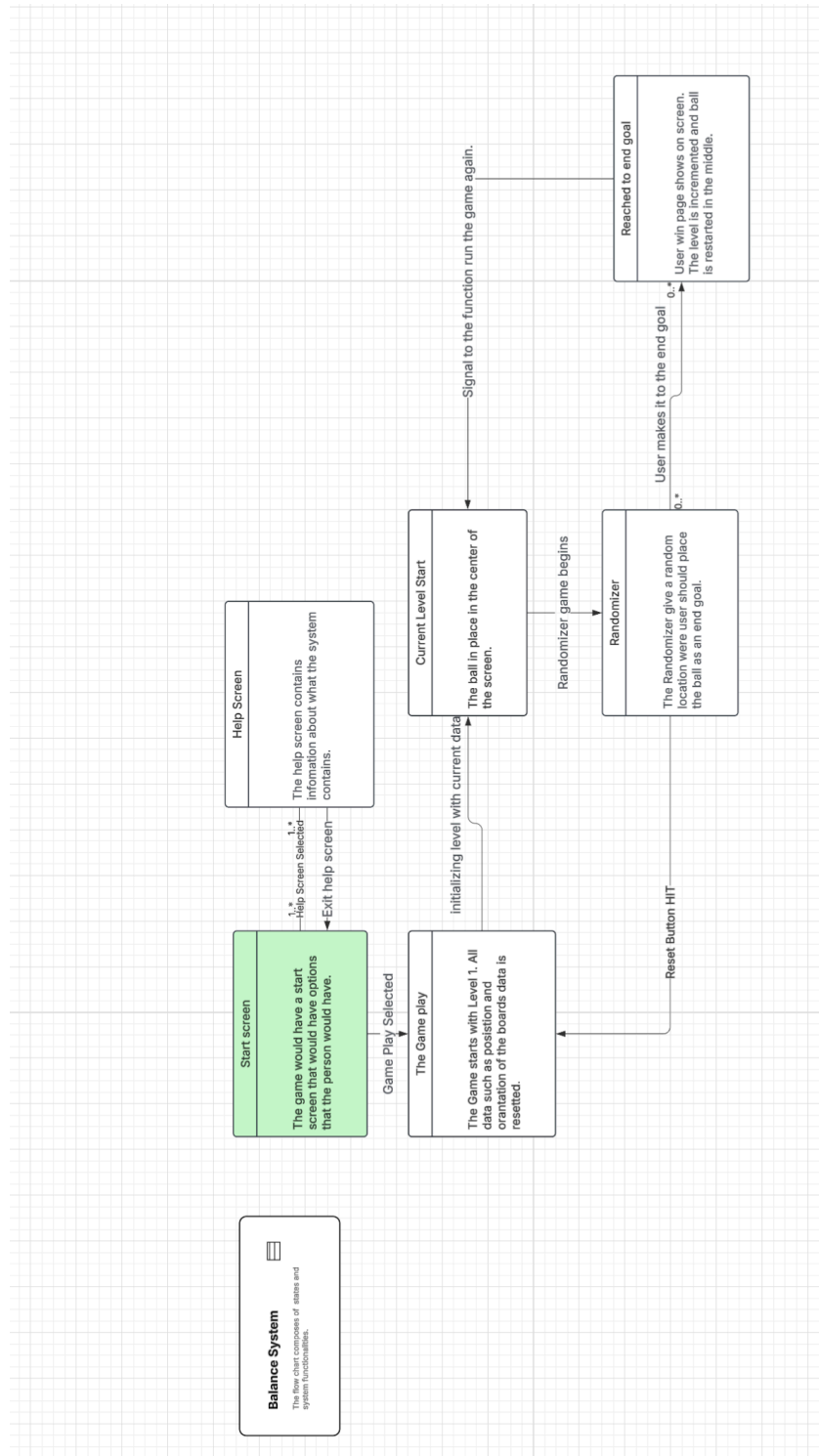


Figure 1: System Overview



1.3 Document Overview

There is no security issue with this device currently. There has not been any testing with the system on a cyber level, currently this system is not harden.



CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
YSM	Yourdon Structured Method
LTDC	Layered Transmission Display Controller
STM	STMicroelectronics
MEMS	Micro-Electro-Mechanical Systems
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:



External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents



CHAPTER 3

Version Description

This chapter provides an output of how version control works.

3.1 Inventory of Materials

ID	Titles	Abrv	Dates	Version	Release	Description
ID1	Buying hardware	BS	09/17/2015	V1.0.0	0	Just purchased the STM32 board

3.2 Inventory of Software

Software Name	Description
Mbed Studio	Deployed code to the STM32 board
Visual Studio Code	Writing the codebase in this editor
Github	Includes Version Source Control and Pipeline Jobs

3.3 Changes From Prior Version

ID	Titles	Abrv	Dates	Version	Release	Description
ID1	Basic setup	BS	09/17/2015	V1.0.0	0	Initial design

3.4 Adaptation Data

The version sets up an system will have a bare minimum approach. Just to have basic functionality.

3.5 Related (Third Party) Documents

Currently there are no third party artifacts that we need for this project.

ID	Titles	Abrv	Dates	Version	Release	Description
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3.6 Installation Instructions

- First connect board to laptop directly through micro-usb.



- Using the embed studio IDE (this might change to use Keli Studio) identify which chip by connecting to it.
- Once the code is up and running then deploy to it.

3.7 Possible Problems and Known Errors

The system is not completely working. There would be problems when running the system.



APPENDIX

Appendix

A.1 Appendix Area 1

This section is ...**TBD**....

A.2 Appendix Area 2

This section is ...**TBD**....



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